

Q-1 Sketch the parametric curve $x = t^2 - 3, y = t + 2, -3 \leq t \leq 3$

Q-2 Sketch the polar curve

a) $r = 1 + 5 \sin \theta$

b) $r = 3 \cos 3\theta$

c) $r = \sin\left(\frac{\theta}{2}\right)$

d) $r^2 = \cos 4\theta$

Q-3 Find the exact length and area of cardioid $r = 2(1 + \cos \theta)$

Q-4 Find the length of

a) $x = e^t - t, y = 4e^{t/2}, 0 \leq t \leq 2$

b) $x = \cos t + \ln(\tan \frac{1}{2}t), y = \sin t, \pi/4 \leq t \leq 3\pi/4.$

c) $\mathbf{r}(t) = \cos t \mathbf{i} + \sin t \mathbf{j} + \ln \cos t \mathbf{k}, 0 \leq t \leq \pi/4$

Q-5 Find the area of given curve

a) enclosed by given parametric curve $x = t^2 - 2t, y = \sqrt{t}$ and y -axis.

b) between the curve $r = 4(1 - \cos \theta), r = 1 + \sin \theta$

c) that lies inside the first and outside the 2nd curve $r = 3 \cos \theta, r = 2 - \cos \theta$

Q-6 Find the unit tangent $\mathbf{T}(t)$, normal vectors $\mathbf{N}(t)$ and the curvature $\mathbf{K}(t)$

a) $\mathbf{r}(t) = \langle \sqrt{2}t, e^t, e^{-t} \rangle$ at $(0, 1, 1)$

b) $\mathbf{r}(t) = \langle \cos t, \sin t, \ln \cos t \rangle, (1, 0, 0)$

Q-7 Find an equation of the tangent to the curve at given point $x = \sqrt{t}, y = t^2 - 2t; t = 4$

Q-8 Find the points on the following curve where the tangents are horizontal or vertical

a) $x = t^3 - 3t, y = t^3 - 3t^2$

b) $r = 1 + \cos \theta$

Q-9 Find parametric equation for tangent line to the curve at given point

$x = \sqrt{t^2 + 3}, y = \ln(t^2 + 3), z = t; (2, \ln 4, 1)$

Q-10 a) Find the exact length of the portion of the curve that lie in first quadrant in figure -1

b). Find the area of shaded region in figure-2

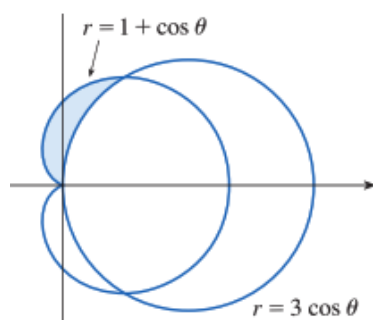


figure-2

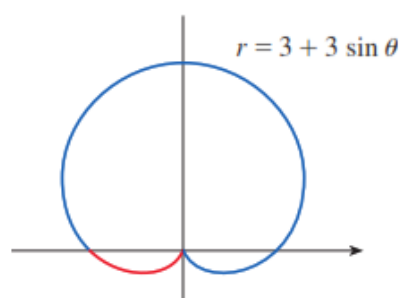


figure-1